LAB -2 DHANUSH.V 1RUA24SCS0031

Lab 2 Ensemble Learning Bagging

In [2]:
```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import BaggingClassifier
from sklearn.metrics import accuracy_score


data = load_iris()
X = data.data
y = data.target


x_train, x_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)


base_model = DecisionTreeClassifier()


bagging_model = BaggingClassifier(
    estimator=base_model,
    n_estimators=50,
    random_state=42
)


bagging_model.fit(x_train, y_train)


y_pred = bagging_model.predict(x_test)


print("Accuracy:", accuracy_score(y_test, y_pred))
```

```
Accuracy: 1.0
```

In [3]:
```
from sklearn.metrics import confusion_matrix
from sklearn.metrics import classification_report
cm = confusion_matrix(y_test, y_pred)
print("Confusion Matrix.\n",cm)
print("\nClassification report:\n", classification_report(y_test, y_pred))
```

```
Confusion Matrix.
 [[19  0  0]
 [ 0 13  0]
 [ 0  0 13]]

Classification report:
               precision    recall  f1-score   support

           0       1.00      1.00      1.00        19
           1       1.00      1.00      1.00        13
           2       1.00      1.00      1.00        13

    accuracy                           1.00        45
   macro avg       1.00      1.00      1.00        45
weighted avg       1.00      1.00      1.00        45
```

In [ ]: